



PK FOKAM INSTITUTE OF EXCELLENCE

Dignity – Faith – Responsibility

DEPARTMENT OF COMPUTER & MECHANICAL ENGINEERING

**Project: Development of a Restaurant Management Web
Application**

REPORT v1.4

COURSE: INTRODUCTION TO PYTHON

THEME: PREPARATION AND RESEARCH FOR THE DEVELOPMENT OF A RESTAURANT WEB
APP

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Work Distribution

The project is carried out by a team of four members. Each member has a specific role in the project while contributing to the tasks required at each

Table 1: General role

Member	Role	Matricule Number
Kamdem Matchum Judith Purchase	Database Analysis, Documentation, Project Coordination	25S052
Mvondo Mvondo Wilfried Anthony	Backend Development	25S048
Massado Djoko Aicha	Frontend Development	26S003
Mep Audrey Larissa	System Integration and Testing	26S014
Ibrahim Diallo	System Analysis and Implementation Support	

Contribution (Current Report)

Since this report covers *the first stage of the project*, the following tasks were completed collaboratively by the team:

Member	Contribution	Percentage
Kamdem Matchum Judith Purchase	Analyzed the SQL database structure, studied most of the system models, prepared the project documentation, and designed the PowerPoint presentation containing the desired GUI	20%
Mvondo Mvondo Wilfried Anthony	Assisted in setting up the Django development environment and project structure, design and corrected the class model and carried out the initial system implementation	20%
Massado Djoko Aicha	Studied the system requirements, Drew and corrected the Use Case Diagram, and participated in the implementation of the welcome page	20%
Mep Meku Audrey	Helped configure development tools, assisted in project planning, and participated in the analysis of the Class Diagram	20%
Ibrahim Dialo	Participated in the study of the system models and assisted in project organization	20%

Table 1.1: Role for stage 1&2

Introduction

The purpose of this project is to design and develop a restaurant management web application that will help restaurants manage their daily operations efficiently. The system will allow restaurant staff to manage

customers, orders, reservations, employees, and inventory through a centralized web interface.

The application will be developed using Python and the Django framework, providing a secure and scalable backend environment. A relational database will store restaurant information such as customer data, menu items, orders, and stock.

This project is carried out as a group assignment and will be completed in three stages, each accompanied by a progress report.

Project Objectives

Main Objective: Develop a functional web-based system that simplifies restaurant management tasks.

Specific objectives:

- Design a database system to manage restaurant data
- Develop backend logic using Django
- Create a user-friendly web interface for staff
- Manage restaurant operations: customer info, orders, reservations, employees, inventory
- Ensure the system is organized, efficient, and easy to use

Project Phases

The development of the Restaurant Management Web Application is divided into three main phases.

Phase 1: Project Initiation (Concept Definition)

This is where the idea of the Restaurant Management Web App is born. The main problem is identified: restaurant operations are often slow and disorganized when handled manually. The solution is defined as a digital system that can centralize and simplify restaurant management tasks using a web application.

Phase 2: Requirements Analysis

In this phase, the system requirements are collected and clarified. This includes defining what the system must do, such as managing users, handling orders, displaying menus, and possibly processing billing. User roles (admin, staff, etc.) are identified, along with functional and non-functional requirements.

Phase 3: System Analysis

The system is broken down logically to understand how it will work internally. Workflows such as ordering, menu updates, and user authentication are analyzed. The goal is to clearly understand system behavior before design begins.

Phase 4: System Design

This phase focuses on planning the structure of the system. UML diagrams such as use case, class, sequence, and activity diagrams are created. The database structure and system architecture are also designed. This phase acts as the blueprint of the entire application.

Phase 5: Development Environment Setup

The development tools and environment are prepared. Python is installed, a Django environment is created (using Anaconda or virtual environment), and Django is installed. The project repository is also set up on GitHub, and the initial project structure is created.

Phase 6: Backend Development

The core logic of the system is implemented using Django. This includes creating models, views, and URL routing. Core features like authentication, menu management, and order processing are developed in this phase.

Phase 7: Database Implementation

The database is designed and connected to the backend system. Models are migrated into actual database tables, and relationships between entities (such as users, orders, and menu items) are established and tested.

Phase 8: Frontend Development

The user interface is developed using HTML, CSS, and possibly Bootstrap. Pages such as login, dashboard, menu, and order interfaces are created to allow users to interact with the system.

Phase 9: System Integration

All components of the system are connected together. The frontend is linked with backend logic and database operations. The full workflow of the system is tested to ensure smooth interaction between all modules.

Phase 10: Testing and Debugging

The system is tested to identify and fix errors. This includes checking individual components (unit testing) and the full system (integration testing). Bugs are corrected and performance is improved.

Phase 11: Deployment

The application is prepared for use. It may be run locally or deployed to a web server. Configuration for production use is done if required.

Phase 12: Documentation and Reporting

All project details are documented, including system design, diagrams, implementation steps, and user instructions. This phase ensures the project can be understood and evaluated by others.

Phase 13: Maintenance and Improvement

After initial completion, the system can be improved by adding new features, fixing remaining issues, or optimizing performance based on feedback.

System Analysis and Description of the Proposed System

The proposed system is a restaurant management web application enabling administrators and staff to manage restaurant operations.

Modules:

- Customer Management: Store and track customer information (Currently working on)
- Order Management: Record and manage customer orders
- Reservation Management: Handle table reservations
- Employee Management: Store employee information
- Inventory and Stock Management: Monitor ingredients and stock levels

All modules interact with a central database for organized and accessible information.

Tools and Technologies

Technology	Type	Purpose
Python	Backend programming language	Used to implement the server-side logic of the restaurant management system
Django Framework	Web development framework	Manages database interactions, web pages, authentication, and application security

Technology	Type	Purpose
MySQL	Database Management System (DBMS)	Stores system data such as customers, orders, products, and employee records
GitHub	Version control platform	Used for source code management, version tracking, and project collaboration
HTML	Frontend technology	Provides the structure of the web pages
CSS	Frontend technology	Provides styling and layout for the user interface

Database Overview

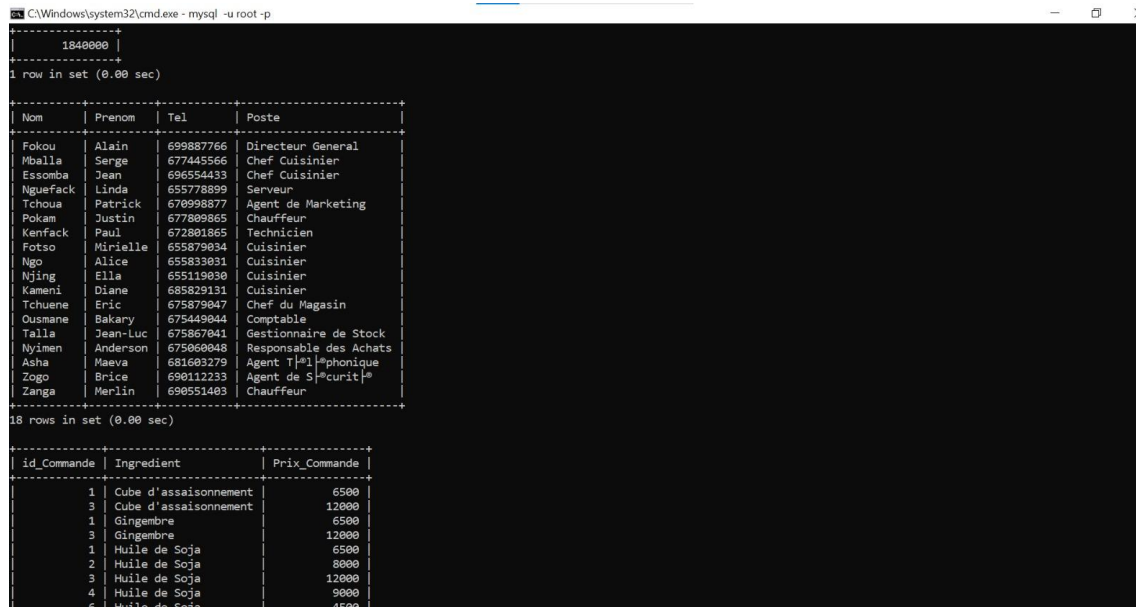
The database stores all information required for restaurant operations.

Tables:

- Customers
- Orders
- Reservations
- Employees
- Products/Menu items

- Stock/Inventory

Sample



```
C:\Windows\system32\cmd.exe - mysql -u root -p
+-----+
| 1840000 |
+-----+
1 row in set (0.00 sec)

+-----+
| Nom      | Prenom | Tel      | Poste      |
+-----+
| Fokou    | Alain  | 699887766 | Directeur General |
| Mballa   | Serge  | 677445566 | Chef Cuisinier |
| Essomba  | Jean   | 696554433 | Chef Cuisinier |
| Nguéfack | Linda  | 655778899 | Serveur |
| Tchoua   | Patrick | 678998877 | Agent de Marketing |
| Pokam    | Justin | 677889865 | Chauffeur |
| Kenfack  | Paul   | 672801865 | Technicien |
| Fotso    | Mirielle | 655879034 | Cuisinier |
| Ngo      | Alice  | 655833031 | Cuisinier |
| Njing    | Ella   | 655119030 | Cuisinier |
| Kameni   | Diane  | 685829131 | Cuisinier |
| Tchuene  | Eric   | 675879047 | Chef du Magasin |
| Ousmane  | Bakary | 675449044 | Comptable |
| Talia    | Jean-Luc | 675867041 | Gestionnaire de Stock |
| Nyimen   | Anderson | 675869048 | Responsable des Achats |
| Asha     | Maeva  | 681683779 | Agent Téléphonique |
| Zogo     | Brice  | 690112233 | Agent de Sécurité |
| Zanga    | Merlin | 690551483 | Chauffeur |
+-----+
18 rows in set (0.00 sec)

+-----+
| id_Commande | Ingredient      | Prix_Commande |
+-----+
| 1 | Cube d'assaisonnement | 6500 |
| 3 | Cube d'assaisonnement | 12000 |
| 1 | Gingembre | 6500 |
| 3 | Gingembre | 12000 |
| 1 | Huile de Soja | 6500 |
| 2 | Huile de Soja | 8000 |
| 3 | Huile de Soja | 12000 |
| 4 | Huile de Soja | 9000 |
| 6 | Huile de Soja | 4500 |
+-----+
```

Figure a: Data in the sql file

The design ensures organized, consistent, and retrievable data. The SQL file is included with this report.

Project Planning:

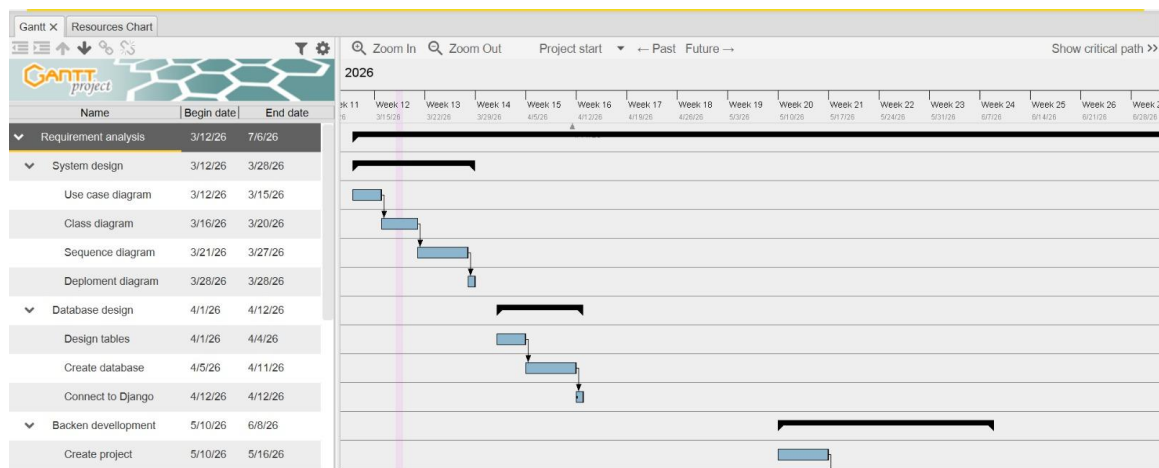


Figure 2: Gantt Chart

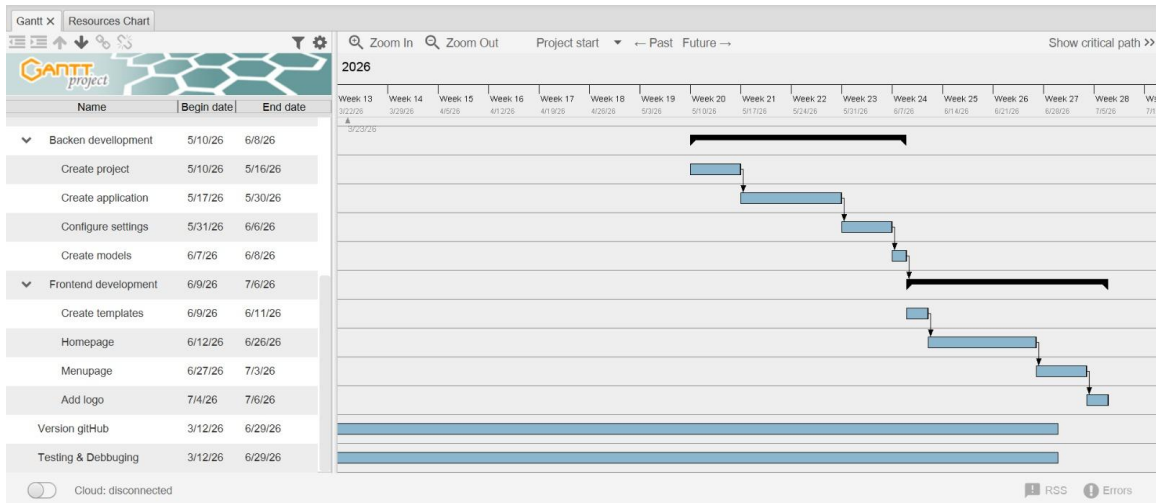


Figure 3: Gantt chart

System Modeling

To better understand the structure and behavior of the system, several UML diagrams were developed.

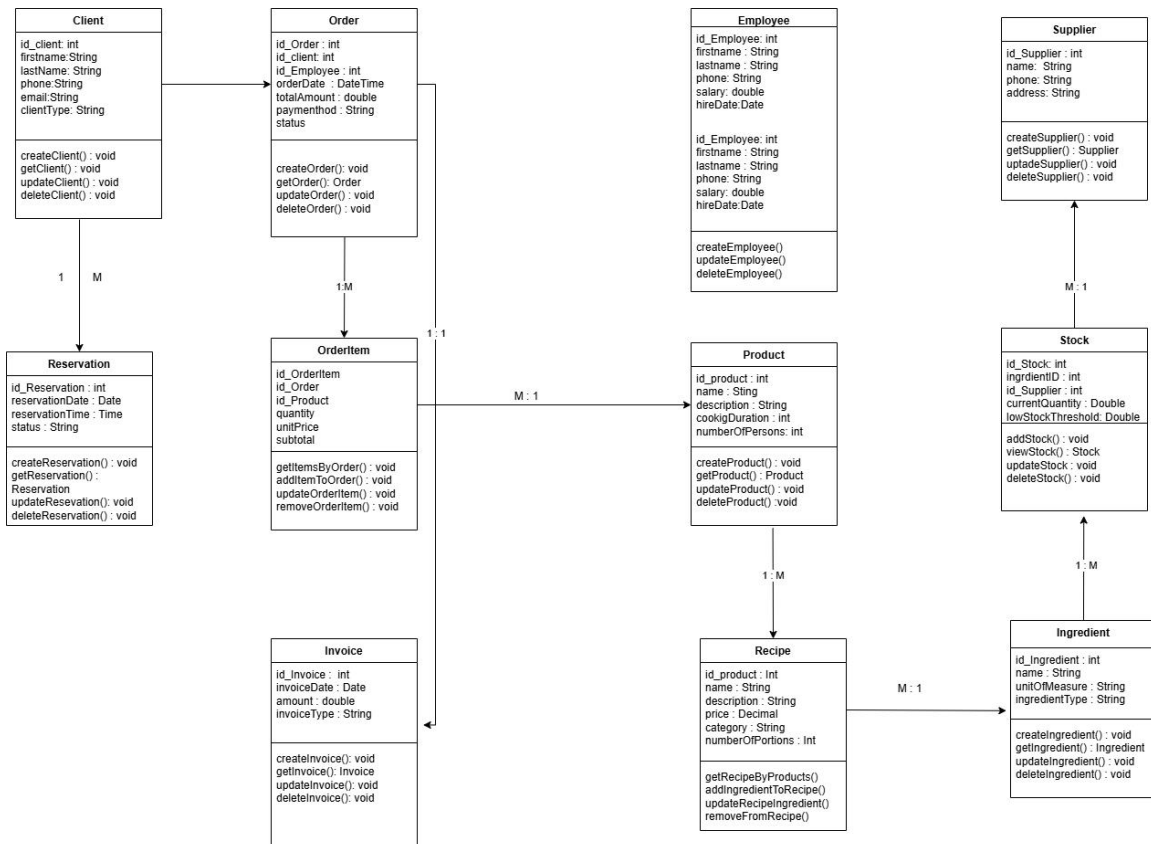


Figure 4: Class model

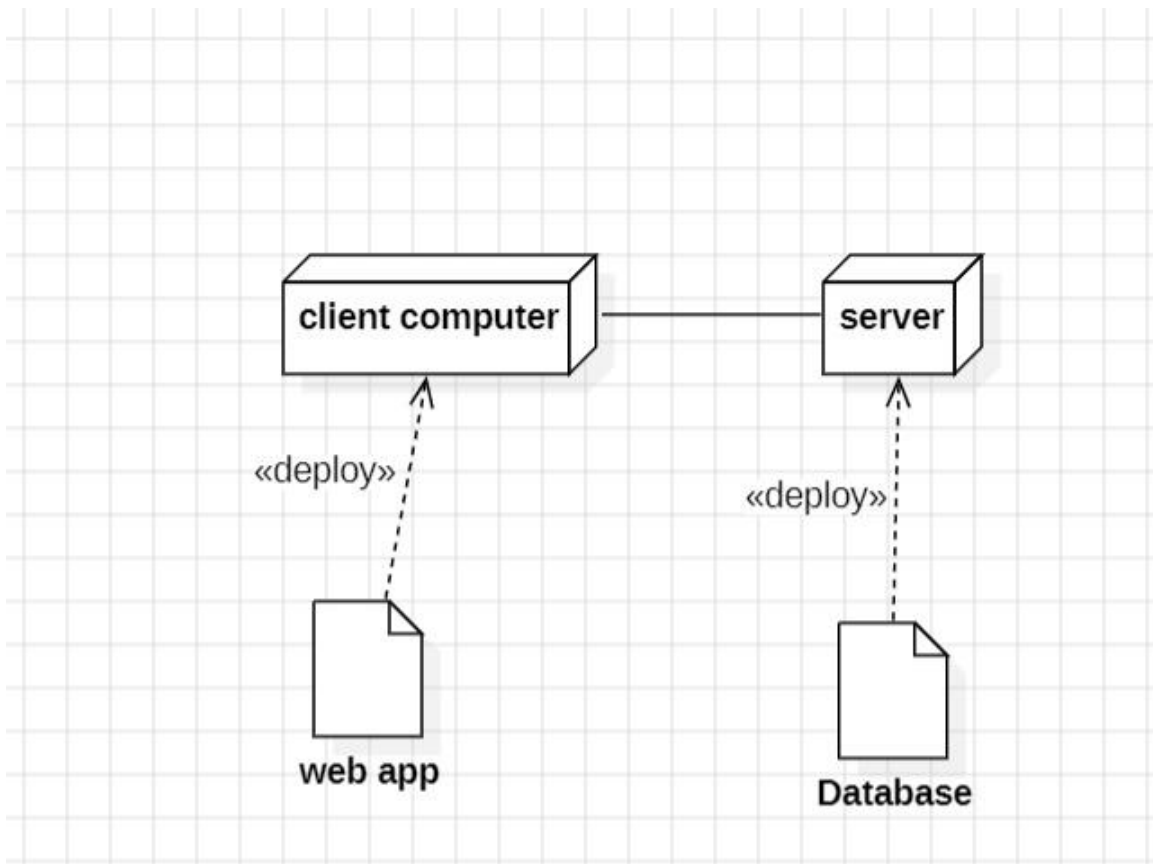


Figure 5: Deployment Diagram

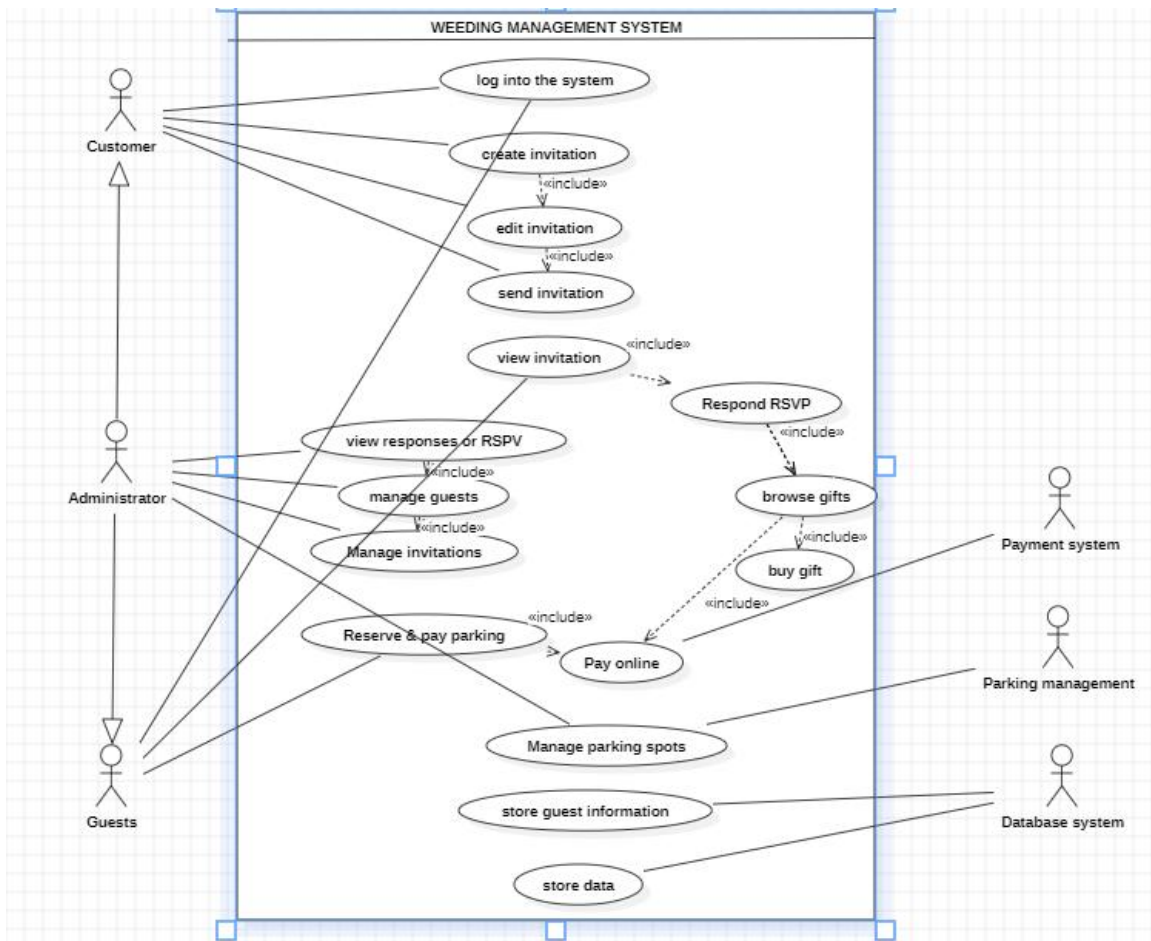


Figure 11: User case diagram

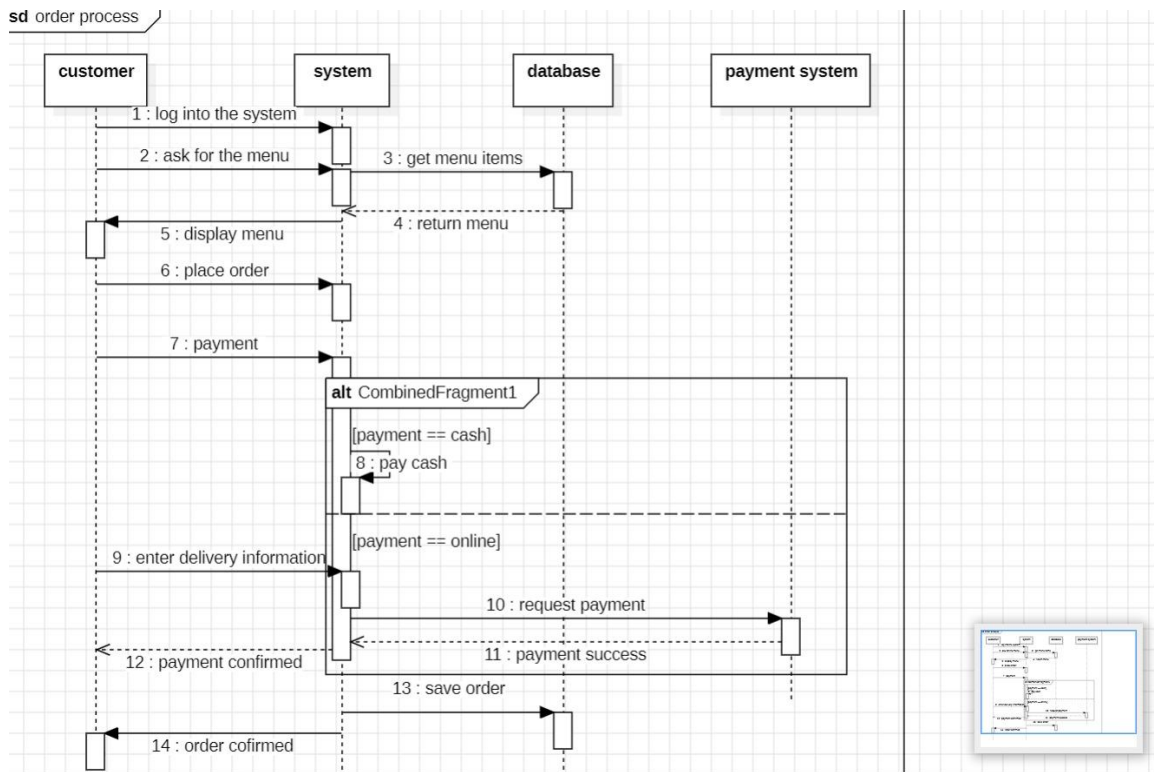


Figure 6: Sequence Diagram

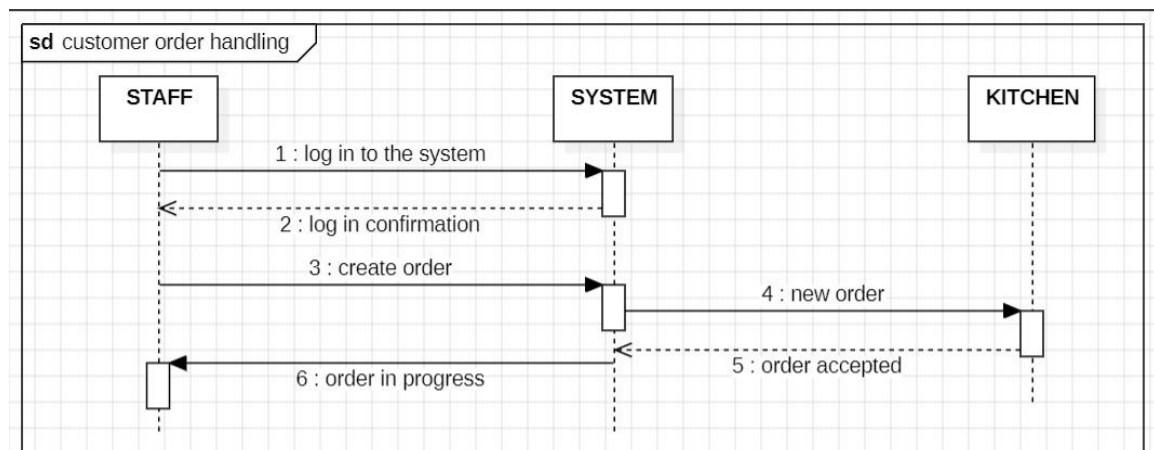


Figure 7: Sequence diagram continuity

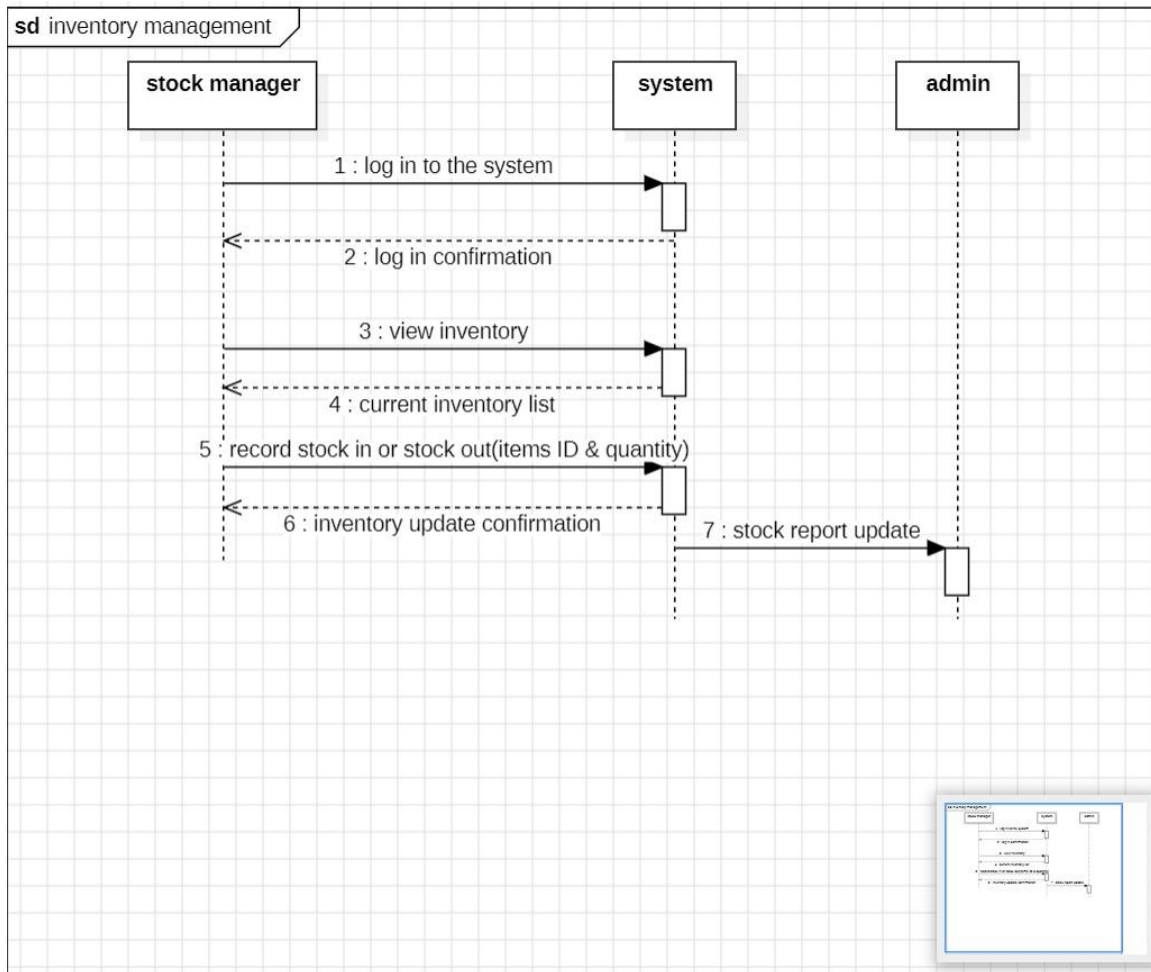


Figure 8: Sequence diagram end

System Implementation

Implementation is the phase where the design of the restaurant management system is transformed into a working application. At this stage, the different modules identified during the analysis and design phases are progressively developed. The objective is to build the backend structure, database, and graphical user interfaces of the system.

Database Implementation

The database was implemented using MySQL based on the class diagram produced during the design phase. The main tables created include:

- Products
- Customers
- Orders
- Reservations
- Employees
- Inventory

The database structure allows the system to store restaurant menu items, customer information, orders, and stock levels.

Backend Implementation

The backend logic is currently being developed using Python and the Django framework. At this stage, CRUD operations (Create, Read, Update, Delete) are being implemented starting with the product management module.

Currently implemented features:

- Add product
- View products
- Modify product
- Delete product

Graphical User Interface Implementation

The graphical user interfaces are being progressively developed starting with the Product Management module. The first interfaces created allow the administrator to add, view, edit and delete products.

Below are screenshots of the first front end implementation results.

Example:

Add /Edit a product

Name:

Description:

Price:

Save

Back

Image 1: Add Product Interface

List of products

Add a product

poulet dg - 2500.00 FCFA	Modify	Delete
fried rice - 1500.00 FCFA	Modify	Delete

Image 2: Product list interface

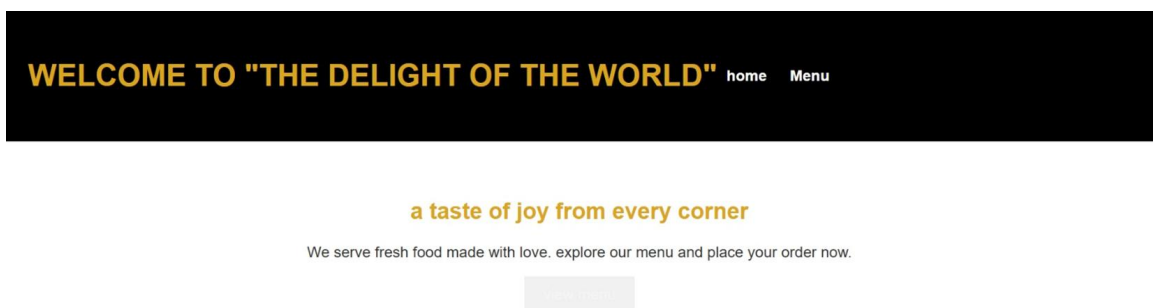


Image 3: Welcome page

This interface allows the administrator to insert new products into the system by providing the name, description, and price.

Overviews of our desired Graphical User Interface

Found in the powerpoint upload in our github repository

Title : GENERAL OVERVIEW OF OUR GRAPICAL USER INTERFACE

Current Progress of the Project

Completed tasks:

- Installed and configured development tools (Python, Django)

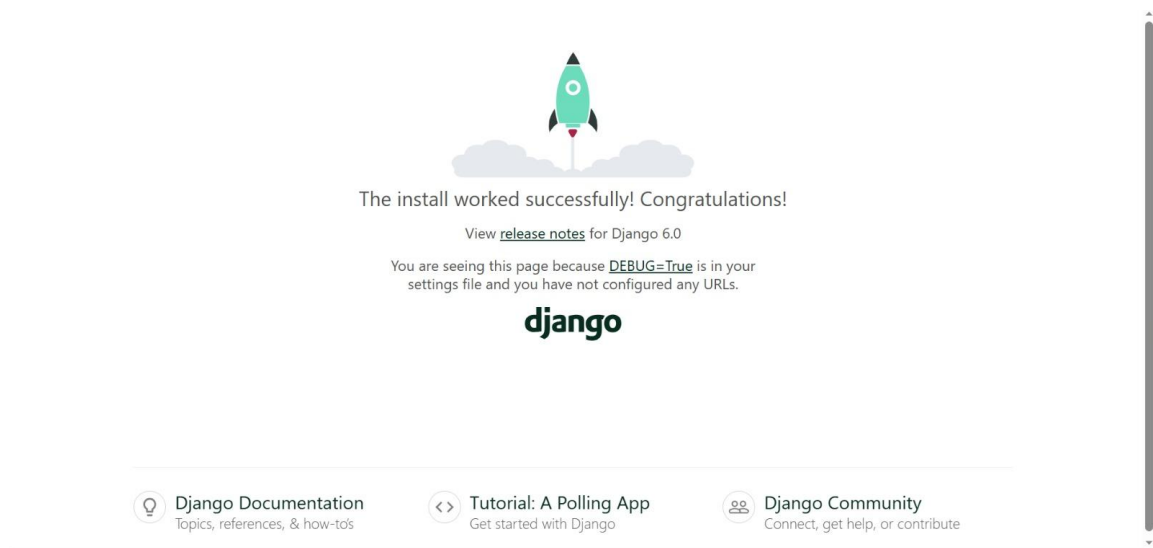


Figure 9: Django complete installation

- Created GitHub repository

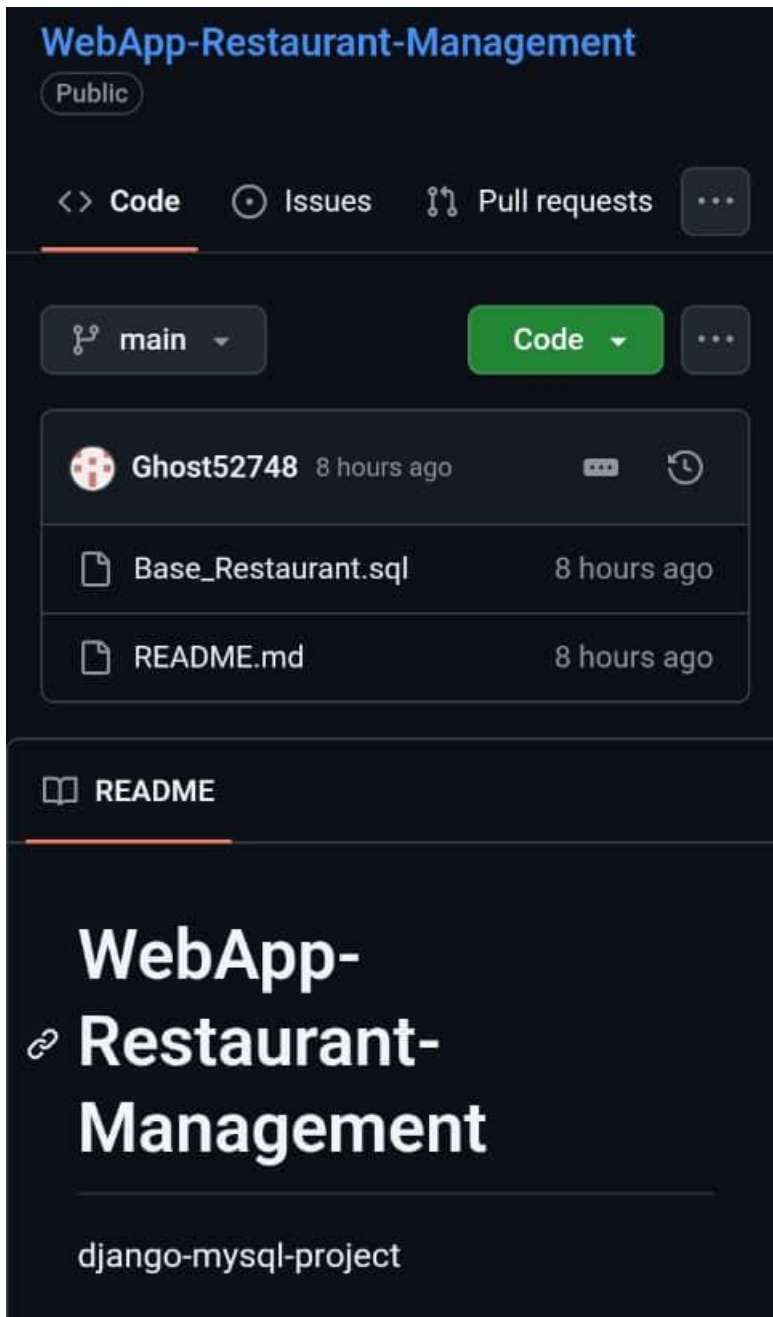


Figure 10: Github repository

- Created Django project structure



Figure 11: Project structure

- Implemented database models in Django

```
C:\Windows\system32\cmd.exe
(venv) C:\Users\admin\django_project\config\config\notepad settings.py

(venv) C:\Users\admin\django_project\config\config>cd ..

(venv) C:\Users\admin\django_project\config>python manage.py dbshell
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 51
Server version: 8.0.40 MySQL Community Server - GPL

Copyright (c) 2000, 2024, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW TABLES
+-----+
| Tables_in_restaurant_db |
+-----+
| auth_group                |
| auth_group_permissions    |
| auth_permission           |
| auth_user                 |
| auth_user_groups          |
| auth_user_user_permissions|
| django_admin_log          |
| django_content_type       |
| django_migrations         |
| django_session            |
+-----+
10 rows in set (0.00 sec)

mysql> exit;
Bye

(venv) C:\Users\admin\django_project\config>python manage.py migrate admin
Operations to perform:
  Apply all migrations: admin
```

Figure 12:Database models in Django

- Connected Django application to the database

Database link with Django

1840000			
1 row in set (0.00 sec)			
Nom	Prenom	Tel	Poste
Fokou	Alain	699887766	Directeur General
Mballa	Serge	677445566	Chef Cuisinier
Essomba	Jean	696554433	Chef Cuisinier
Nguefack	Linda	655778899	Serveur
Tchoua	Patrick	670998877	Agent de Marketing
Pokam	Justin	677809865	Chauffeur
Kenfack	Paul	672801865	Technicien
Fotso	Mirielie	655879034	Cuisinier
Ngo	Alice	655833031	Cuisinier
Njing	Ella	655119030	Cuisinier
Kameni	Diane	685829131	Cuisinier
Tchuene	Eric	675879047	Chef du Magasin
Ousmane	Bakary	675449044	Comptable
Talla	Jean-Luc	675867041	Gestionnaire de Stock
Nyimen	Anderson	675060048	Responsable des Achats
Asha	Maeva	681603279	Agent T ^l phonique
Zogo	Brice	690112233	Agent de S ^l curit ^e
Zanga	Merlin	690551403	Chauffeur
18 rows in set (0.00 sec)			
id_Commande	Ingredient	Prix_Commande	
1	Cube d'assaisonnement	6500	
3	Cube d'assaisonnement	12000	
1	Gingembre	6500	
3	Gingembre	12000	
1	Huile de Soja	6500	
2	Huile de Soja	8000	
3	Huile de Soja	12000	
4	Huile de Soja	9000	
6	Huile de Soja	4500	

Figure 13: Database link with Django

- Project plan using gantt chart

- Understanding and representing the management system using models
- Analyze SQL database structure and table relationships
- Study Django framework more deeply

In progress:

- Continuous improvement of system architecture and project plan
- Implementation of the Product management module

Difficulties Encountered

During the first phase of the project, the group encountered several technical and organizational challenges:

- Learning the Django framework for some members who were not previously familiar with it
- Understanding the database structure due to the presence of multiple related tables
- Difficulty understanding and creating UML diagrams such as Class, Use Case, and Sequence diagrams
- Installing and configuring development tools such as Python, Django, StarUML, and GanttProject
- Coordinating tasks and planning work effectively among group members
- Understanding how to properly analyze system requirements before designing the models
- Connecting the Django application to the MySQL database
- Retrieving data from HTML forms caused issues
- Errors occurred during update and delete operations due to mistakes in SQL queries and route configuration.

Solutions Applied

To overcome these challenges, the group adopted the following solutions:

- Members studied Django documentation, watched tutorials, and practiced small examples to improve their understanding of the framework
 - The group analyzed the SQL database structure together to better understand relationships between tables and how data flows in the system
 - Members researched UML modeling concepts and collaborated to better understand how to design Class, Use Case, and Sequence diagrams
 - Members who successfully installed the required tools assisted others in configuring their development environment
 - Tasks were clearly divided among members according to roles and competencies to improve efficiency
 - Regular communication and group discussions were used to clarify requirements and improve project planning
-
- SQL queries were reviewed and debugged to ensure that update and delete operations functioned correctly.
 - CSS styling was gradually added to improve the interface layout and usability.

Next Steps and Implementation Plan

1. Implement core functionalities: orders, customers, reservations
2. Connecting all modules to the CSS design

3. Implementing customer management
4. Implementing order processing
5. Developing reservation management
6. Improving dashboard interface
7. System testing
8. Testing and integration of modules

GitHub Repository Link

<https://github.com/Ghost52748/WebApp-Restaurant-Management>

SQL Database File

The SQL file containing the database structure is included in our Github repository as part of the project submission.


```

DROP DATABASE IF EXISTS Restaurant;
CREATE DATABASE Restaurant CHARACTER SET utf8mb4 COLLATE utf8mb4_general_ci;
USE Restaurant;
/* ===== TABLES ===== */
CREATE TABLE Client (
  id_Client INT PRIMARY KEY,
  Nom VARCHAR(20),
  Prenom VARCHAR(20),
  Tel VARCHAR(15),
  Email VARCHAR(45),
  Type_de_Client VARCHAR(20)
);
CREATE TABLE `Table` (
  id_Table INT PRIMARY KEY,
  Numero_Table VARCHAR(3),
  Capacite INT,
  Commentaire TEXT
);
CREATE TABLE Reservation (
  id_Reservation INT PRIMARY KEY,
  id_Client INT,
  id_Table INT,
  `Date` DATE,
  Heure TIME,
  Statut VARCHAR(45),
  FOREIGN KEY (id_Client) REFERENCES Client(id_Client),
  FOREIGN KEY (id_Table) REFERENCES `Table`(id_Table)
);
CREATE TABLE Ingredient (
  id_Ingredient INT PRIMARY KEY,
  Nom VARCHAR(45),
  Unite_de_Mesure VARCHAR(45),
  `Type` VARCHAR(30)
);

```

Figure 14: Database file